CYBERSECURITY INTERNSHIP REPORT

Task 27: Backup Security

Track: Cyber Security **Organization:** Veda Technology

1. Introduction

Backup security involves creating and protecting copies of important data so it can be restored after accidental deletion, hardware failure, malware, ransomware, or other incidents.

2. Objective

To design a basic backup strategy for a small organization and understand backup and recovery security.

3. Proposed Backup Strategy

Identify important data: Include documents, databases, employee records, financial files, and configuration files.
Follow the 3-2-1 rule: Maintain three copies of data, use two different storage types, and keep one copy off-site.
Schedule backups: Run daily incremental backups and weekly full backups.
Encrypt data: Protect backup data both in transit and at rest.
Restrict access: Allow only authorized personnel to access or restore backups.
Protect against ransomware: Maintain an offline or immutable copy.
Monitor and test: Review backup logs and perform regular restoration tests.
Set retention: Keep backups according to business needs and applicable policies.

4. Example Backup Schedule

- Daily: Incremental backup of important files.
- Weekly: Full backup to a separate storage location.
- Monthly: Maintain an encrypted off-site copy and review retention.
- Quarterly: Conduct a recovery test and document the result.

5. The 3-2-1 Backup Rule

3 copies: Keep the original data plus two backup copies.
2 storage types: Store copies on two different types of storage media.
1 off-site copy: Keep one copy in a separate physical location or secure remote environment.

6. Why Test Backups?

Testing confirms that data can be restored correctly. It helps identify corrupted backups, missing files, permission issues, and recovery problems before an actual incident occurs.

7. Recommended Security Controls

Use encryption, multi-factor authentication, least-privilege access, offline or immutable backups, monitoring of backup logs, and documented restoration procedures.

8. Conclusion

A secure backup strategy helps an organization protect important information and recover from data loss, system failures, and cyberattacks. Applying the 3-2-1 rule, encrypting backups, restricting access, and testing restoration regularly improves data availability and business continuity.